

# What you'll build:

# Customer service multi-agent system

## Introduction

### Putting it all together

In modules 1-3, you learned the building blocks:

- 1 Part 1: Multi-agent systems and `sub_agents` hierarchy
- 2 Part 2: Workflow agents (Sequential, Parallel, Loop)
- 3 Part 3: Agent communication with `output_key` and `{key}` templating

Now you'll combine all three into one complete application.

This capstone also integrates features from earlier lessons in the learning path. You'll see how tools (lesson 3), callbacks (lesson 5), state namespaces (lesson 4), and content safety patterns (lesson 6) work together with multi-agent orchestration to create production-ready systems.



# What you'll build:

## Customer service multi-agent system

A complete customer service application with:

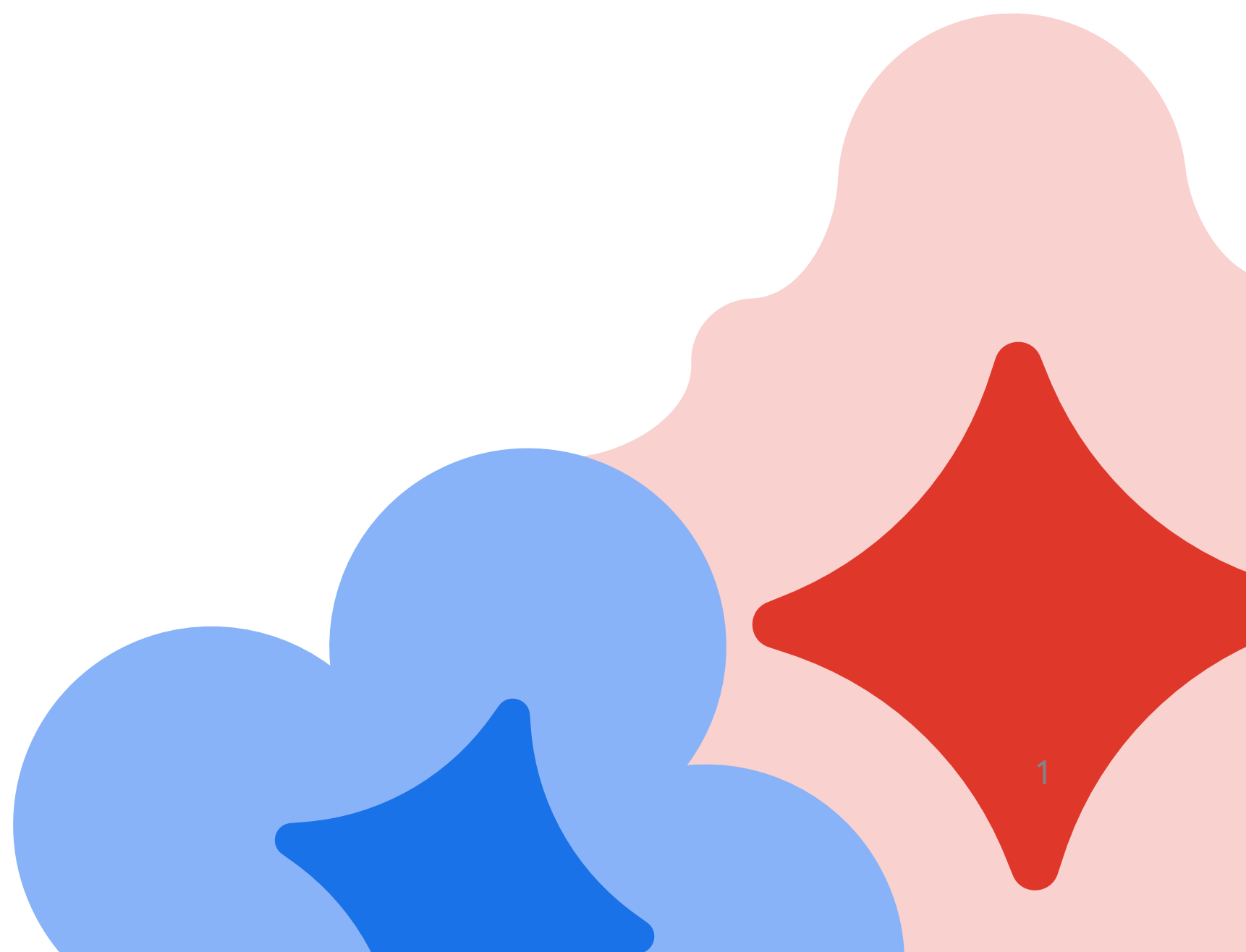
- 1 Intake agent - Extracts ticket details from user request
- 2 Parallel specialists - Billing and technical experts run together
- 3 Response agent - Creates final response using all information

### Architecture:

```
None
User Request
  ↓
Intake Agent (extracts details)
  ↓
Parallel Agents (billing + technical research)
  ↓
Response Agent (creates final answer)
```

### Concepts used:

- SequentialAgent (ordered execution)
- ParallelAgent (concurrent specialists)
- **output\_key** and **{key}** (state communication)
- Multiple LlmAgents working together





The following diagram illustrates the fundamental difference between local development and cloud deployment in terms of accessibility and persistence:

None

```
graph TB
  USER[User Request] --> INTAKE[Intake Agent]
  INTAKE -->|"ticket_info"| BILLING[Billing Researcher]
  INTAKE -->|"ticket_info"| TECH[Technical Researcher]

  subgraph "Parallel Stage"
    BILLING
    TECH
  end

  BILLING -->|"billing_findings"| RESPONSE[Response Agent]
  TECH -->|"technical_findings"| RESPONSE

  RESPONSE --> OUTPUT[Final Response]

  style INTAKE fill:#e1f5ff
  style BILLING fill:#eeffee
  style TECH fill:#eeffee
  style RESPONSE fill:#fff3cd
```

The pipeline flows from intake (blue) through parallel research (green) to final response (yellow).

